

Frame family — scaffolding test

1. `\quadframeleft` — params 1,3: left column x-terms only

$$2x^2 + 25x + 33$$

$$\begin{array}{r} 2x \\ \times \\ x \end{array}$$

$$2x^2 - 25x + 33$$

$$\begin{array}{r} 2x \\ \times \\ x \end{array}$$

$$3x^2 + 26x + 35$$

$$\begin{array}{r} 3x \\ \times \\ x \end{array}$$

$$3x^2 - 26x + 35$$

$$\begin{array}{r} 3x \\ \times \\ x \end{array}$$

2. `\quadframeright` — params 2,4: right column constants only

$$2x^2 + 25x + 33$$

$$\begin{array}{r} + \quad 3 \\ \times \\ + \quad 11 \end{array}$$

$$2x^2 - 25x + 33$$

$$\begin{array}{r} - \quad 3 \\ \times \\ - \quad 11 \end{array}$$

$$3x^2 + 26x + 35$$

$$\begin{array}{r} + \quad 5 \\ \times \\ + \quad 7 \end{array}$$

$$3x^2 - 26x + 35$$

$$\begin{array}{r} - \quad 5 \\ \times \\ - \quad 7 \end{array}$$

3. `\quadframetop` — params 1,2: top row only

$$\begin{array}{ccc}
 2x^2 + 25x + 33 & & \\
 2x & + & 3 \\
 & \diagdown & \diagup \\
 & \diagup & \diagdown
 \end{array}$$

$$\begin{array}{ccc}
 2x^2 - 25x + 33 & & \\
 2x & - & 3 \\
 & \diagdown & \diagup \\
 & \diagup & \diagdown
 \end{array}$$

$$\begin{array}{ccc}
 3x^2 + 26x + 35 & & \\
 3x & + & 5 \\
 & \diagdown & \diagup \\
 & \diagup & \diagdown
 \end{array}$$

$$\begin{array}{ccc}
 3x^2 - 26x + 35 & & \\
 3x & - & 5 \\
 & \diagdown & \diagup \\
 & \diagup & \diagdown
 \end{array}$$

4. `\quadframebottom` — params 3,4: bottom row only

$$\begin{array}{ccc}
 2x^2 + 25x + 33 & & \\
 & \diagdown & \diagup \\
 x & + & 11
 \end{array}$$

$$\begin{array}{ccc}
 2x^2 - 25x + 33 & & \\
 & \diagdown & \diagup \\
 x & - & 11
 \end{array}$$

$$\begin{array}{ccc}
 3x^2 + 26x + 35 & & \\
 & \diagdown & \diagup \\
 x & + & 7
 \end{array}$$

$$\begin{array}{ccc}
 3x^2 - 26x + 35 & & \\
 & \diagdown & \diagup \\
 x & - & 7
 \end{array}$$

5. `\quadframerising` — params 3,2: ends of rising diagonal

$$\begin{array}{ccc}
 2x^2 + 25x + 33 & & \\
 & + & 3 \\
 & \diagdown & \diagup \\
 x & &
 \end{array}$$

$$\begin{array}{ccc}
 2x^2 - 25x + 33 & & \\
 & - & 3 \\
 & \diagdown & \diagup \\
 x & &
 \end{array}$$

$$3x^2 + 26x + 35$$

$$\begin{array}{c} + \\ x \quad 5 \end{array}$$

$$3x^2 - 26x + 35$$

$$\begin{array}{c} - \\ x \quad 5 \end{array}$$

6. \quadframefalling — params 1,4: ends of falling diagonal

$$2x^2 + 25x + 33$$

$$\begin{array}{c} 2x \\ + \\ 11 \end{array}$$

$$2x^2 - 25x + 33$$

$$\begin{array}{c} 2x \\ - \\ 11 \end{array}$$

$$3x^2 + 26x + 35$$

$$\begin{array}{c} 3x \\ + \\ 7 \end{array}$$

$$3x^2 - 26x + 35$$

$$\begin{array}{c} 3x \\ - \\ 7 \end{array}$$

7. \quadframebare — all four entries; student does working

$$2x^2 + 25x + 33$$

$$\begin{array}{c} + \\ 2x \quad 3 \\ + \\ x \quad 11 \end{array}$$

$$2x^2 - 25x + 33$$

$$\begin{array}{c} - \\ 2x \quad 3 \\ - \\ x \quad 11 \end{array}$$

$$3x^2 + 26x + 35$$

$$\begin{array}{ccc} 3x & + & 5 \\ & \times & \\ x & + & 7 \end{array}$$

$$3x^2 - 26x + 35$$

$$\begin{array}{ccc} 3x & - & 5 \\ & \times & \\ x & - & 7 \end{array}$$

8. `\quadframe` — full worked frame

$$\begin{array}{ccc} 2x^2 + 25x + 33 \\ 2x & + & 3 \\ & \times & \\ x & + & 11 \end{array} \left| + \right. \frac{3x}{25x} \checkmark$$

$$\begin{array}{ccc} 2x^2 - 25x + 33 \\ 2x & - & 3 \\ & \times & \\ x & - & 11 \end{array} \left| + \right. \frac{3x}{25x} \checkmark$$

$$\begin{array}{ccc} 3x^2 + 26x + 35 \\ 3x & + & 5 \\ & \times & \\ x & + & 7 \end{array} \left| + \right. \frac{5x}{26x} \checkmark$$

$$\begin{array}{ccc} 3x^2 - 26x + 35 \\ 3x & - & 5 \\ & \times & \\ x & - & 7 \end{array} \left| + \right. \frac{5x}{26x} \checkmark$$

9. `\quadframesign` — full frame with sign arrows

$$\begin{array}{ccc} 2x^2 & \overset{+}{\circlearrowleft} & 25x + 33 \\ 2x & + & 3 \\ & \times & \\ x & + & 11 \end{array} \left| + \right. \frac{3x}{25x} \checkmark$$

$$\begin{array}{ccc} 2x^2 & \overset{-}{\circlearrowleft} & 25x + 33 \\ 2x & - & 3 \\ & \times & \\ x & - & 11 \end{array} \left| + \right. \frac{3x}{25x} \checkmark$$

$$\begin{array}{r}
 3x^2 \quad \textcircled{+} 26x + 35 \\
 \swarrow \quad \searrow \\
 3x \quad + \quad 5 \\
 \quad \quad \quad \downarrow \\
 x \quad + \quad 7
 \end{array}
 \left| \begin{array}{r}
 5x \\
 + \\
 \frac{21x}{26x} \quad \checkmark
 \end{array} \right.$$

$$\begin{array}{r}
 3x^2 \quad \textcircled{-} 26x + 35 \\
 \swarrow \quad \searrow \\
 3x \quad - \quad 5 \\
 \quad \quad \quad \downarrow \\
 x \quad - \quad 7
 \end{array}
 \left| \begin{array}{r}
 5x \\
 + \\
 \frac{21x}{26x} \quad \checkmark
 \end{array} \right.$$

Bracket family — scaffolding test

10. \quadfirst — first terms of each factor shown

$$\begin{array}{l}
 2x^2 + 25x + 33 \\
 (2x \quad)(x \quad)
 \end{array}$$

$$\begin{array}{l}
 2x^2 - 25x + 33 \\
 (2x \quad)(x \quad)
 \end{array}$$

$$\begin{array}{l}
 3x^2 + 26x + 35 \\
 (3x \quad)(x \quad)
 \end{array}$$

$$\begin{array}{l}
 3x^2 - 26x + 35 \\
 (3x \quad)(x \quad)
 \end{array}$$

11. \quadlast — last terms of each factor shown

$$\begin{array}{l}
 2x^2 + 25x + 33 \\
 (\quad + 3)(\quad + 11)
 \end{array}$$

$$\begin{array}{l}
 2x^2 - 25x + 33 \\
 (\quad - 3)(\quad - 11)
 \end{array}$$

$$\begin{array}{l}
 3x^2 + 26x + 35 \\
 (\quad + 5)(\quad + 7)
 \end{array}$$

$$\begin{array}{l}
 3x^2 - 26x + 35 \\
 (\quad - 5)(\quad - 7)
 \end{array}$$

12. \quadouter — outer terms: first of first, last of last

$$\begin{array}{l}
 2x^2 + 25x + 33 \\
 (2x \quad)(\quad + 11)
 \end{array}$$

$$\begin{array}{l}
 2x^2 - 25x + 33 \\
 (2x \quad)(\quad - 11)
 \end{array}$$

$$\begin{array}{l}
 3x^2 + 26x + 35 \\
 (3x \quad)(\quad + 7)
 \end{array}$$

$$\begin{array}{l}
 3x^2 - 26x + 35 \\
 (3x \quad)(\quad - 7)
 \end{array}$$

13. \quadinner — inner terms: last of first, first of last

$$\begin{array}{l}
 2x^2 + 25x + 33 \\
 (\quad + 3)(x \quad)
 \end{array}$$

$$\begin{array}{l}
 2x^2 - 25x + 33 \\
 (\quad - 3)(x \quad)
 \end{array}$$

$$3x^2 + 26x + 35$$

$$(\quad + 5)(x \quad)$$

$$3x^2 - 26x + 35$$

$$(\quad - 5)(x \quad)$$

14. \texttt{\backslashquadfactors} — both factors complete

$$2x^2 + 25x + 33$$

$$(2x + 3)(x + 11)$$

$$2x^2 - 25x + 33$$

$$(2x - 3)(x - 11)$$

$$3x^2 + 26x + 35$$

$$(3x + 5)(x + 7)$$

$$3x^2 - 26x + 35$$

$$(3x - 5)(x - 7)$$

15. \texttt{\backslashquadroots} — factors and roots

$$2x^2 + 25x + 33$$

$$(2x + 3)(x + 11) = 0$$

$$x = (3)(2)\frac{0}{0} \text{ or } x = (11)(1)\frac{0}{0}$$

$$2x^2 - 25x + 33$$

$$(2x - 3)(x - 11) = 0$$

$$x = (-3)(2)\frac{0}{0} \text{ or } x = (-11)(1)\frac{0}{0}$$

$$3x^2 + 26x + 35$$

$$(3x + 5)(x + 7) = 0$$

$$x = (5)(3)\frac{0}{0} \text{ or } x = (7)(1)\frac{0}{0}$$

$$3x^2 - 26x + 35$$

$$(3x - 5)(x - 7) = 0$$

$$x = (-5)(3)\frac{0}{0} \text{ or } x = (-7)(1)\frac{0}{0}$$

Factor each expression.

16. $x^2 + 3x + 2$

$x^2 + 4x + 3$

$x^2 + 6x + 5$

$x^2 + 8x + 7$

17. $x^2 + x - 2$

$x^2 + 4x - 5$

$x^2 + 6x - 7$

$x^2 + 10x - 11$

18. $x^2 + 2x - 3$

$x^2 + 10x - 11$

$x^2 + 4x - 5$

$x^2 + 6x - 7$

19. $x^2 - x - 2$

$x^2 - 2x - 3$

$x^2 - 4x - 5$

$x^2 - 6x - 7$

20. $x^2 - 3x + 2$

$x^2 - 4x + 3$

$x^2 - 6x + 5$

$x^2 - 8x + 7$

- | | | | | |
|-----|-------------------|-------------------|--------------------|---------------------|
| 21. | $x^2 + 13x + 22$ | $x^2 + 7x + 10$ | $x^2 + 33x + 62$ | $x^2 + 26x + 69$ |
| 22. | $x^2 + 5x - 66$ | $x^2 - 3x - 130$ | $x^2 - 17x - 110$ | $x^2 + 8x - 105$ |
| 23. | $x^2 - x - 30$ | $x^2 - x - 42$ | $x^2 - x - 56$ | $x^2 - x - 72$ |
| 24. | $x^2 - 2x - 35$ | $x^2 - 2x - 63$ | $x^2 - 2x - 99$ | $x^2 - 2x - 143$ |
| 25. | $x^2 - 3x - 40$ | $x^2 - 3x - 70$ | $x^2 - 3x - 108$ | $x^2 - 3x - 154$ |
| 26. | $x^2 - 5x - 50$ | $x^2 - 5x - 150$ | $x^2 - 5x - 300$ | $x^2 - 5x - 500$ |
| 27. | $2x^2 + 7x + 3$ | $5x^2 + 36x + 7$ | $3x^2 + 34x + 11$ | $7x^2 + 15x + 2$ |
| 28. | $21x^2 + 10x + 1$ | $10x^2 + 7x + 1$ | $33x^2 + 14x + 1$ | $65x^2 + 18x + 1$ |
| 29. | $2x^2 + 5x + 3$ | $5x^2 + 12x + 7$ | $3x^2 + 14x + 11$ | $7x^2 + 9x + 2$ |
| 30. | $2x^2 + 5x - 3$ | $5x^2 - 34x - 7$ | $3x^2 + 32x - 11$ | $7x^2 - 13x - 2$ |
| 31. | $2x^2 - x - 3$ | $5x^2 + 2x - 7$ | $3x^2 - 8x - 11$ | $7x^2 - 5x - 2$ |
| 32. | $2x^2 + 13x + 15$ | $3x^2 + 26x + 35$ | $5x^2 + 62x + 77$ | $7x^2 + 102x + 143$ |
| 33. | $2x^2 + 7x - 15$ | $3x^2 - 16x - 35$ | $5x^2 + 48x - 77$ | $7x^2 - 80x - 143$ |
| 34. | $2x^2 + 21x + 27$ | $3x^2 + 10x + 8$ | $3x^2 + 40x + 125$ | $8x^2 + 43x + 15$ |
| 35. | $2x^2 + 15x - 27$ | $3x^2 + 2x - 8$ | $3x^2 - 4x - 64$ | $8x^2 + 43x + 15$ |

Factoring frame — worked examples

36.

$$\begin{array}{r|l}
 \begin{array}{r} 2x^2 \\ 2x \\ x \end{array} \begin{array}{r} + \\ + \\ + \end{array} \begin{array}{r} 3 \\ 11 \end{array} & \begin{array}{l} \textcircled{+} 25x + 33 \\ + \\ \hline \frac{22x}{25x} \end{array} \checkmark
 \end{array}$$

$$\begin{array}{r|l}
 \begin{array}{r} 2x^2 \\ 2x \\ x \end{array} \begin{array}{r} - \\ - \\ - \end{array} \begin{array}{r} 3 \\ 11 \end{array} & \begin{array}{l} \textcircled{-} 25x + 33 \\ + \\ \hline \frac{22x}{25x} \end{array} \checkmark
 \end{array}$$

$$\begin{array}{r|l}
 \begin{array}{r} 2x^2 \\ 2x \\ x \end{array} \begin{array}{r} + \\ - \\ - \end{array} \begin{array}{r} 11 \\ 5 \end{array} & \begin{array}{l} \textcircled{+} 1x - -55 \\ - \\ \hline \frac{10x}{x} \end{array} \checkmark
 \end{array}$$

$$\begin{array}{r|l}
 \begin{array}{r} 3x^2 \\ 3x \\ x \end{array} \begin{array}{r} - \\ + \\ + \end{array} \begin{array}{r} 5 \\ 2 \end{array} & \begin{array}{l} \textcircled{+} 1x - -10 \\ - \\ \hline \frac{6x}{x} \end{array} \checkmark
 \end{array}$$

$$\begin{array}{r|l}
 \begin{array}{r} 2x^2 \\ 2x \\ x \end{array} \begin{array}{r} + \\ + \\ + \end{array} \begin{array}{r} 11 \\ 5 \end{array} & \begin{array}{l} \textcircled{+} 21x + 55 \\ + \\ \hline \frac{10x}{21x} \end{array} \checkmark
 \end{array}$$

$$\begin{array}{r|l}
 \begin{array}{r} 3x^2 \\ 3x \\ x \end{array} \begin{array}{r} + \\ + \\ + \end{array} \begin{array}{r} 5 \\ 2 \end{array} & \begin{array}{l} \textcircled{+} 11x + 10 \\ + \\ \hline \frac{6x}{11x} \end{array} \checkmark
 \end{array}$$

$$437x^2 + 1566x + 649$$

$$\begin{array}{r} 23x \quad + \quad 11 \\ \quad \times \quad \quad \quad \\ 19x \quad + \quad 59 \end{array} \bigg| + \begin{array}{r} 209x \\ \hline 1566x \end{array} \checkmark$$

$$2x^2 + 25x + 33$$

$$\begin{array}{r} 2x \quad + \quad 3 \\ \quad \times \quad \quad \quad \\ x \quad + \quad 11 \end{array}$$

$$3x^2 - 11x + 10$$

$$\begin{array}{r} 3x \quad - \quad 5 \\ \quad \times \quad \quad \quad \\ x \quad - \quad 2 \end{array}$$